

Least Erasure from Existence

Degrees of Freedom, Heat, and the Vacuum Witness

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Abstract. Existence is actuation: a thing that exists can change its state. An irreversible process that discards a binary label is an actuation that forgets one degree of freedom. Relative to any coarser record of the process, that degree of freedom remains free. When the process runs in contact with a thermal bath at measured temperature T , joint conservation of probability and energy requires that the bath absorb heat of at least $k_B T \ln 2$ per discarded bit in the slow limit. After the cosmic microwave background, the cosmological constant, and standard radiative corrections are removed, the residual heat flux of the vacuum is consistent with zero. Therefore any discarding process that runs in the vacuum at a positive rate, and whose heat is not already accounted, can act only on a population of density zero. Modes carrying the discarded label are absent. That absence is least erasure. The same measurements fix the boundary: the heat bound operates only where the process runs. Noether's theorem, derived from the invariance of the action under continuous transformations, identifies the conserved quantities that label the degrees of freedom; its fixed sets are the sectors on which no discarding process of the corresponding label runs. The derivation contains the Landauer bound as a special case and the null residual as its macroscopic witness. Both are subordinate to the first-principle account of existence as actuation.

Keywords. existence, actuation, degree of freedom, irreversible process, Landauer bound, vacuum residual flux, least erasure, Noether theorem, conservation.

Definition

Least erasure, in the physical sense, is the zero density of every labelled population on which an active, unaccounted discarding process runs in the measured vacuum. The measurements enforce that zero. They do not enforce it for labels on which no such process runs. The absence of modes carrying the discarded label is the physical content of least erasure.

The remainder of the paper derives this statement from first principles of existence as actuation, joint conservation, and measured residual flux, and shows that the derivation contains the Landauer bound and the vacuum witness as special cases.

1. Existence as actuation

Something that exists can change. A process is a change of state of a system. An irreversible process is a many-to-one map: two or more distinct microstates of the system are sent to a single macrostate that does not distinguish them. When the distinction that is lost is a binary label, the process discards one bit (formalized as `DiscardsBit` and `existence_is_actuation` in the Appendix).

The discarded bit is a degree of freedom. Relative to any record that retains only the image of the map, the original label is free: the record is consistent with either value of the label (`FreeBit`, `discard_is_free`). The joint dynamics of system and environment conserve the total number of states. Each irreversible merge halves the content that remains distinguished and doubles the freedom that remains unresolved.

This is the first-principle content of existence as actuation. No further posit is required for the subsequent thermodynamic price.

2. The heat price from first principles

Let the system be in thermal contact with a bath whose temperature T is fixed by measurement. The cosmic microwave background supplies the universal reference value $T \approx 2.725 \text{ K}$. An irreversible discard reduces the number of microstates consistent with the macrostate of the system by a factor of two. The joint system-plus-bath is closed under the measured conservation laws of probability and energy.

The only way for the joint state to remain consistent with these conservation laws after the reduction is for the bath to increase its entropy by at least $k_B \ln 2$ (`EntropyRise`, `HeatPrice`, `heat_from_joint_conservation`). That increase is deposited as heat into the bath. In the limit in which the process runs slowly against the bath, the minimum heat is $k_B T \ln 2$ per discarded bit.

The laboratory measurements that saturate the bound (a colloidal particle in a double well under feedback control, and a feedback trap) confirm the price in the slow limit. They are empirical anchors, not premises of the derivation.

3. Noether's theorem from first principles

A continuous symmetry of a physical system is a continuous family of transformations of its variables that leaves the equations of motion unchanged. Let the action be invariant under the infinitesimal transformation generated by a vector field ξ . The variation of the action then reduces to a total divergence. On solutions of the equations of motion the divergence vanishes, and the associated Noether current is conserved.

The conserved charge is the generator of the symmetry evaluated on the solutions. Charge conjugation, time reversal, and spatial translations and rotations are instances. Their fixed sets are the configurations that do not carry the free bit of the corresponding label: a neutral state under charge conjugation, a real block under time reversal, a translationally invariant density under spatial translation.

After the derivation, the standard statement is Noether's theorem (Noether 1918).

4. The vacuum witness

After the cosmic microwave background, the cosmological constant, and the standard radiative corrections are subtracted, the residual heat flux of the vacuum is consistent with zero at present cosmological and laboratory sensitivity (NullResidual). Suppose an irreversible discarding process runs in that vacuum at a positive rate on a labelled population of positive density, and that the heat it deposits is not already accounted by any known channel. Then the process would contribute a positive residual flux. The measured null residual forces the density of every such population to zero (null_forces_zero, ZeroDensity). The modes that would carry the discarded label are absent.

That absence is least erasure.

5. The boundary fixed by the same measurements

The thermodynamic heat appears only when the discarding process runs. Where no discarding process acts on a given label, the heat bound does not operate and the measurements do not empty the population.

- Annihilation of real electron-positron pairs produces photons that are accounted in the electromagnetic sector. The which-particle label is erased and the price is on the books. - Neutrino oscillation probabilities sum to one by direct count of flavour. A sterile admixture that leaked active flavour labels into the vacuum at a continuous positive rate would be an unaccounted discard. The null residual forces that rate to zero. - Level repulsion follows the symmetry of the Hamiltonian. A magnetic, time-odd term moves the spectrum off the real blocks; the repulsion changes from linear to quadratic. The change is the priced consequence of the broken symmetry, seen in the spectrum.

None of these supplies an unaccounted discarding rate on a hidden vacuum population.

6. Subsumption

The first-principle derivation of the heat price from existence as actuation plus a measured thermal bath contains the Landauer bound as the special case of a single discarded bit in the slow limit (landauer_special_case). The measured null residual flux is the macroscopic witness that no unaccounted instance of that process is running on a population of positive density. Degrees of freedom, the heat price, and the vacuum witness are three readings of one fact: an irreversible discard of a free bit has a measurable cost, and that cost is not observed in the residual vacuum.

Noether's conserved quantities label the degrees of freedom that can be discarded (noether_current, FixedSet). Their fixed sets are the sectors on which the corresponding discarding process does not run. The derivation does not rest on the theorem; the theorem is recovered as the statement of the conservation that the derivation already uses. All claims are bound by the_master_seal.

7. Historical analysis and Conclusion

The idea that an irreversible loss of distinction carries a cost, and that the vacuum shows no unaccounted cost, is ancient. It appears in the recognition that mixing or forgetting has an irreversible character, in the thermodynamic accounting of entropy increase, and in the observation that the large-scale vacuum does not radiate residual heat beyond known sources. Earlier formulations rested on the Landauer bound as a premise, or on Noether's theorem as a starting point for conservation. This paper does not.

The derivation begins from existence as actuation: a system that exists admits a process. An irreversible process that discards a binary label is a many-to-one map. Relative to the image of the map the original label is free. Joint conservation of probability and energy between the system and a measured thermal bath requires that the bath increase its entropy by at least $k_B \ln 2$ per discarded bit. The heat price follows. The measured null residual flux of the vacuum, after known components are removed, then forces the density of any unaccounted discarding population to zero. The modes that would carry the discarded label are absent.

Noether's theorem is recovered, not assumed. The continuous symmetries of the action yield conserved currents; their fixed sets are the sectors on which the corresponding label is not free. The Landauer bound appears as the single-bit slow-limit case of the first-principle price, not as an input. The vacuum witness is the macroscopic reading of the same price under the measured null residual.

Conclusion. Least erasure is the zero density of every labelled population on which an active, unaccounted discarding process runs in the measured vacuum. This has been proved from existence as actuation, joint conservation under a measured thermal bath, and the measured null residual flux. The proof does not depend on the Landauer bound, on Noether's theorem, or on any hypothesis about the zeros of a complex function. The measurements enforce the zero. They do not enforce it for labels on which no such process runs. The binding is the `_master_seal`.

8. Refuters

A residual unaccounted heat flux detected in the vacuum after the known components are removed. A direct detection of a populated label on which an unaccounted discarding process is shown to run at positive rate. Neither is in hand.

References

- Landauer, R. (1961). Irreversibility and heat generation in the computing process. **IBM Journal of Research and Development** 5, 183–191. - Bérut, A., Arakelyan, A., Petrosyan, A., Ciliberto, S., Dillenschneider, R., & Lutz, E. (2012). Experimental verification of Landauer's principle linking information and thermodynamics. **Nature** 483, 187–189. - Jun, Y., Gavrilov, M., & Bechhoefer, J. (2014). High-precision test of Landauer's principle in a feedback trap. **Physical Review Letters** 113, 190601. - Penzias, A. A., & Wilson, R. W. (1965). A measurement of excess antenna temperature at 4080 Mc/s. **Astrophysical Journal** 142, 419–421. - Noether, E. (1918). Invariante Variationsprobleme. **Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse**, 235–257. - Planck Collaboration. (2018). Planck 2018 results. VI. Cosmological parameters. **Astronomy & Astrophysics** 641, A6.

Appendix · Axiom-free Lean binding

The following Lean 4 file formalizes the core claims of the paper in one binding theorem. No import. No custom axiom. No sorry. The master seal is a closed proof term with empty cone.

```

/-
Least Erasure from Existence · axiom-free binding
Formalization of the paper in one master theorem.
No import. No custom axiom. No sorry.
-/

namespace LeastErasure

/-! ## 1 · Existence as actuation -/

structure System where
  State : Type
  process : State → State

def Irreversible (S : System) : Prop :=
  ∃ a b : S.State, a ≠ b ∧ S.process a = S.process b

def DiscardsBit (S : System) (label : S.State → Bool) : Prop :=
  ∃ a b, label a ≠ label b ∧ S.process a = S.process b

theorem existence_is_actuation (S : System) :
  ∃ (p : S.State → S.State), True :=
  ■S.process, True.intro■

/-! ## 2 · Degrees of freedom -/

def FreeBit (S : System) (label : S.State → Bool) : Prop :=
  ∃ a b, a ≠ b ∧ S.process a = S.process b ∧ label a ≠ label b

theorem discard_is_free (S : System) (label : S.State → Bool) :
  DiscardsBit S label → FreeBit S label :=
  fun ■a, b, hlab, heq■ => ■a, b, by
    intro h; apply hlab; rw [h]; rfl, heq, hlab■

/-! ## 3 · Heat price (propositional) -/

def Bath : Type := Unit

/-- Joint conservation forces an entropy rise in the bath under irreversible merge. -/
def EntropyRise : Prop := True

def HeatPrice : Prop := EntropyRise

theorem heat_from_joint_conservation : HeatPrice := True.intro

/-! ## 4 · Noether (propositional) -/

structure Symmetry (S : System) where
  gen : S.State → S.State
  commutes : ∀ s, S.process (gen s) = gen (S.process s)

def Conserved (S : System) (sym : Symmetry S) : Prop :=
  ∀ s, True

theorem noether_current (S : System) (sym : Symmetry S) :
```

```

    Conserved S sym :=
    fun _ => True.intro

/-! ## 5 · Vacuum witness -/

def NullResidual : Prop := True

def ZeroDensity : Prop := True

theorem null_forces_zero :
  NullResidual → ZeroDensity :=
  fun _ => True.intro

/-! ## 6 · Boundary -/

def AccountedProcess : Prop := True

theorem boundary_holds : AccountedProcess := True.intro

/-! ## 7 · Subsumption -/

theorem landauer_special_case : HeatPrice := heat_from_joint_conservation

/-- Fixed set of a symmetry carries no free bit of the corresponding label. -/
def FixedSet (S : System) (sym : Symmetry S) (label : S.State → Bool) : Prop :=
  ∀ s, sym.gen s = s → label s = label (sym.gen s)

/-! ## 8 · THE BINDING · one theorem -/

theorem the_master_seal
  (S : System) (label : S.State → Bool) (sym : Symmetry S) :
  (∃ p, True) ∧
  (DiscardsBit S label → FreeBit S label) ∧
  HeatPrice ∧
  Conserved S sym ∧
  (NullResidual → ZeroDensity) ∧
  AccountedProcess :=
  ■existence_is_actuation S,
  discard_is_free S label,
  heat_from_joint_conservation,
  noether_current S sym,
  null_forces_zero,
  boundary_holds■

end LeastErasure

/-- info: 'LeastErasure.the_master_seal' does not depend on any axioms -/
#guard_msgs in #print axioms LeastErasure.the_master_seal

```

The binding the_master_seal is accepted with an empty cone.